

Lecture 1: Overview of GIS and ArcGIS

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Two objectives

- Learn how to design an offline mobile mapping system using ArcGIS collector
- Learn how to map parcels using ArcGIS collector and manage the collected data

Topics

- Day 1

- Morning

- Overview of GIS and its capabilities
- Different tools in ArcGIS: What they can do and what you need to know
- ArcOnline

- Afternoon

- ArcOnline details: managing data
- ArcOnline details: Creating basemaps and features for field data collection
- ArcCollector basics
- Arc Collector Mapping and Syncing

- Day 2
- Morning
 - Transferring files and Managing Collected Maps
 - Field Testing
- Afternoon
 - Recaps

Focus

- Creating ArcGIS Collector basemaps
- Data collection using ArcGIS Collector
- Managing data



Why GIS (Geographical Information System)

- Location matters, context matters
- GIS allows relating information that would otherwise be disparate
 - E.g., soil quality information, crops grown, temperature, distance to road, market location

Discussion questions

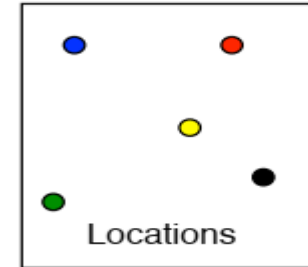
- How many farmers grow tobacco? Where?
- How can we know about this?
- Any other questions we should explore based on your work

GIS process as a data process

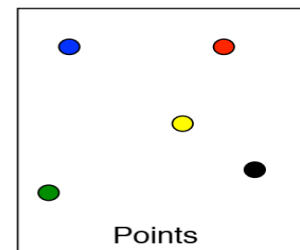
- Planning
 - What's the goal
 - Who will design the system, collect data and manage data
- Data collection
 - Who, when, where, how
- Data management and analysis
 - Who, when, where, how
- Information sharing
 - Who, when, where, how
- Archiving
 - Who, when, where, how

Types of data

- Spatial data
 - Describes the location of geographic features
- Tabular data
 - Describes the characteristics of spatial features
 - Also known as attribute data
 - Example: spreadsheets
- Combine attribute and coordinates



ID	Color
1	Blue
2	Red
3	Green
4	Yellow
5	Black



ID	Color	Use
1	Blue	Hospital
2	Red	Fire Dept.
3	Green	Office
4	Yellow	University
5	Black	Retail

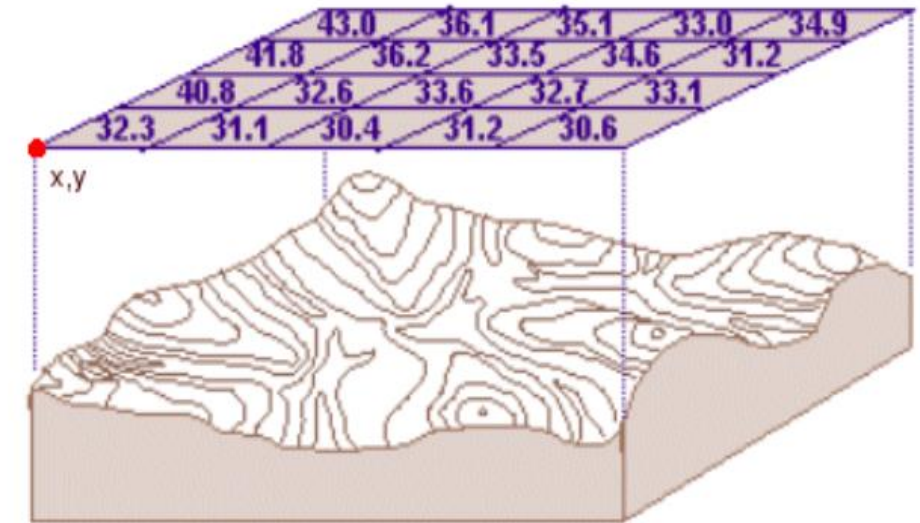
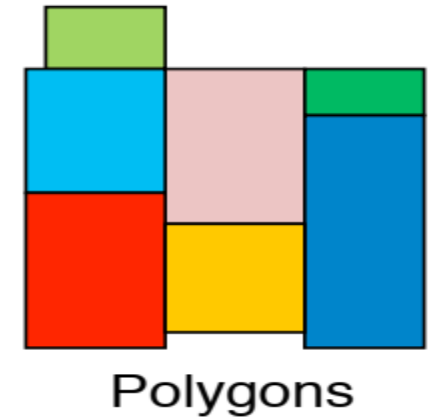
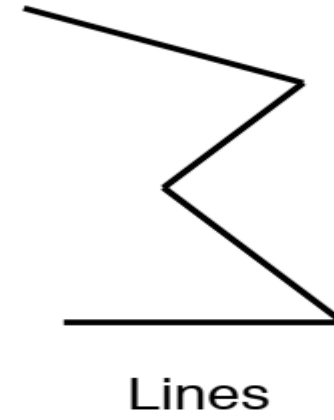
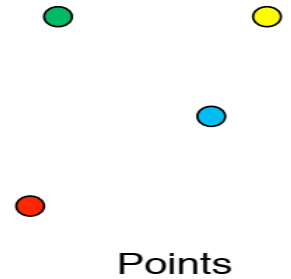
Types of spatial data

- Vector

- **Points:** single location on earth
 - Lat-long, UTM coordinates
 - Example: Address of a farmer's house
- **Lines:** string of points joined by line segments.
 - Has length
 - Examples: Road, river
- **Polygon:** Closed line representing a defined area on the earth.
 - Has perimeter and area. Example: farm parcel

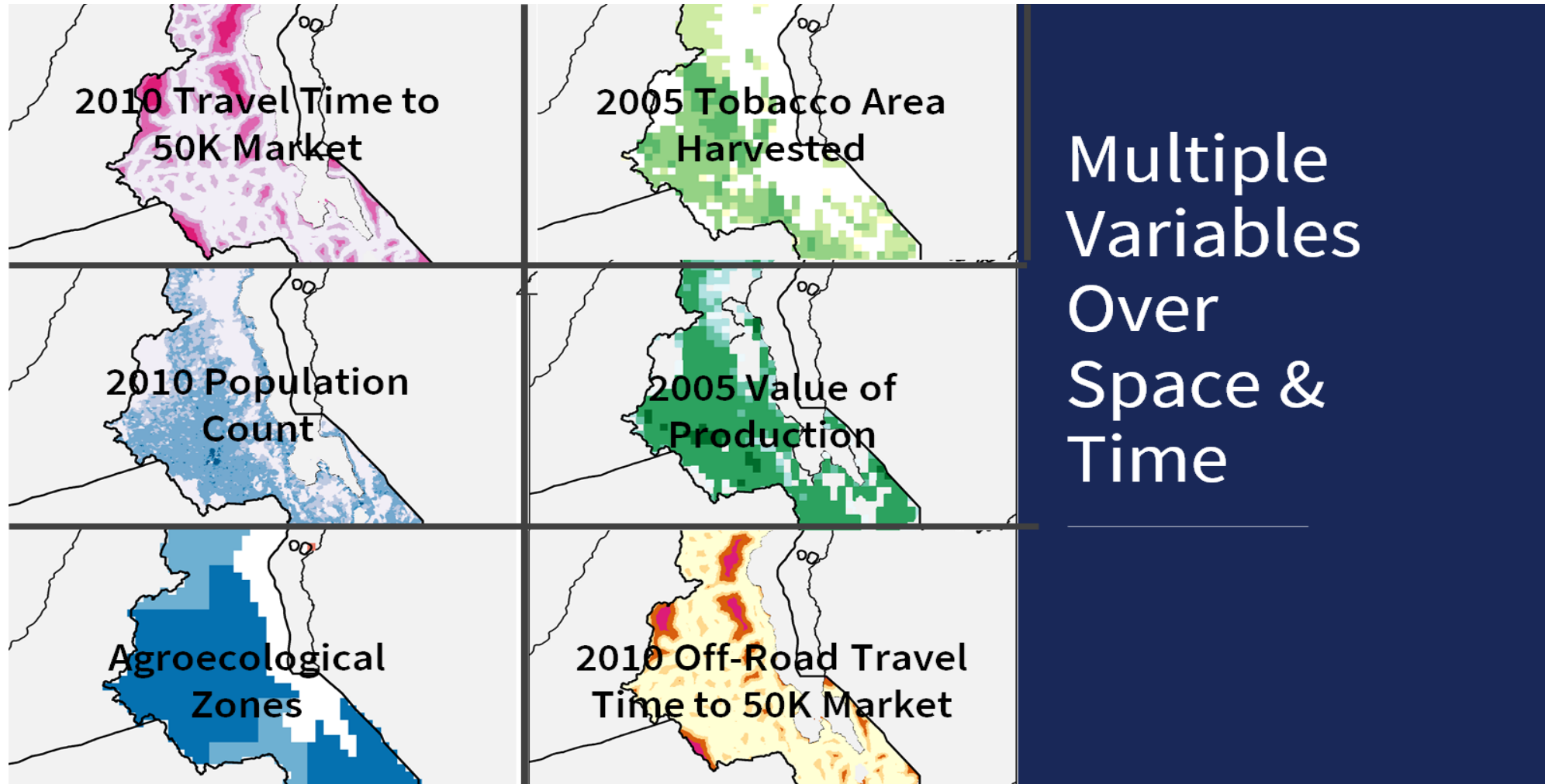
- Raster

- Composed of a regular grid of cells
- Every grid cell has a value
- Every point of the ground belongs to a grid cell
- Examples: Elevation, temperature



AVERAGE ELEVATION

Examples: Raster



Spatial data formats to know

- Shapefile
 - Most common spatial data format in GIS
 - For vector data: points, lines, or polygons
- Web Map Services
- GPS Exchange Format
- Keyhole Markup Language (Google KML)

Spatial data sources in Malawi

- Malawi Spatial Data Platform (<http://www.masdap.mw/>)

Common Coordinate Reference Systems (CRS)

- WGS 84 (World Geodetic system)
- UTM (Universal Transverse Mercator)

GIS Software

- ArcGIS
- QGIS
- R
- POSTGIS
- Many others

ArcGIS suite

- ArcGIS Desktop
 - ArcMap
 - Arc Catalog (e.g., creating file geodatabase)
 - Arc Tools (e.g., spatial analyst tool to extract raster data using vector map)
- ArcGIS Online
- Mobile ArcGIS
 - ArcGIS Collector
 - Survey 123

ArcOnline: Web Mapping Basics

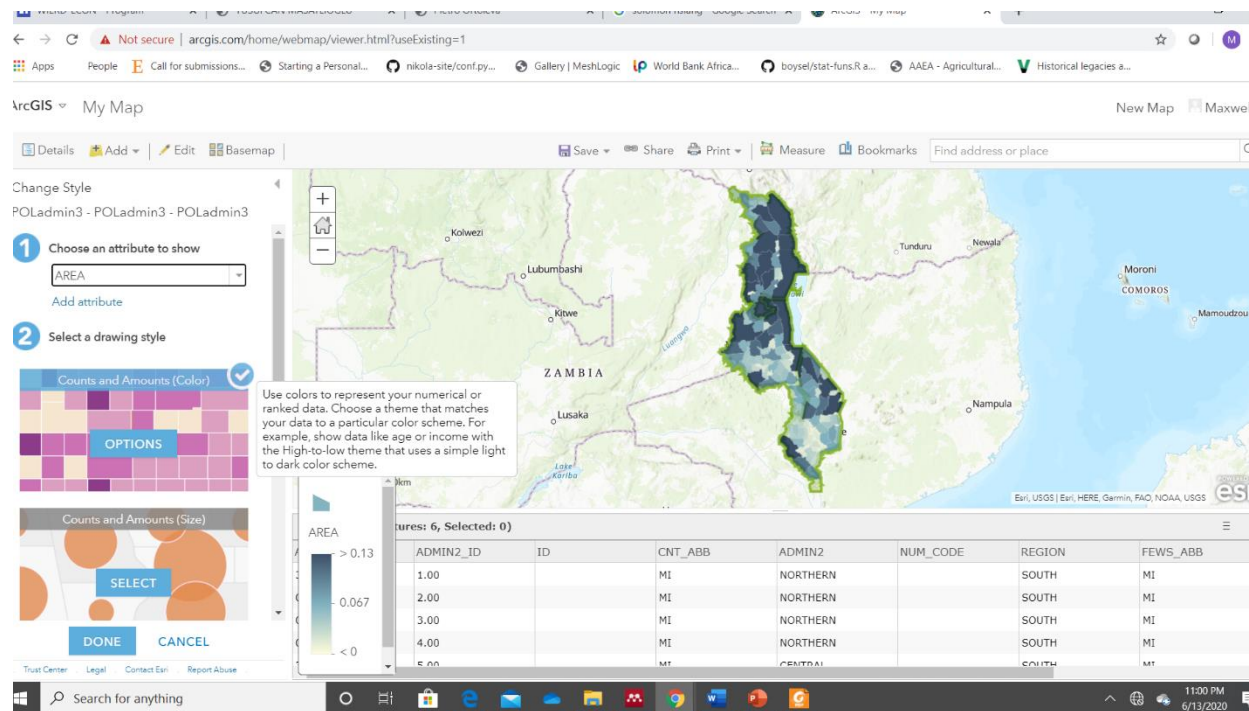
Why web maps?

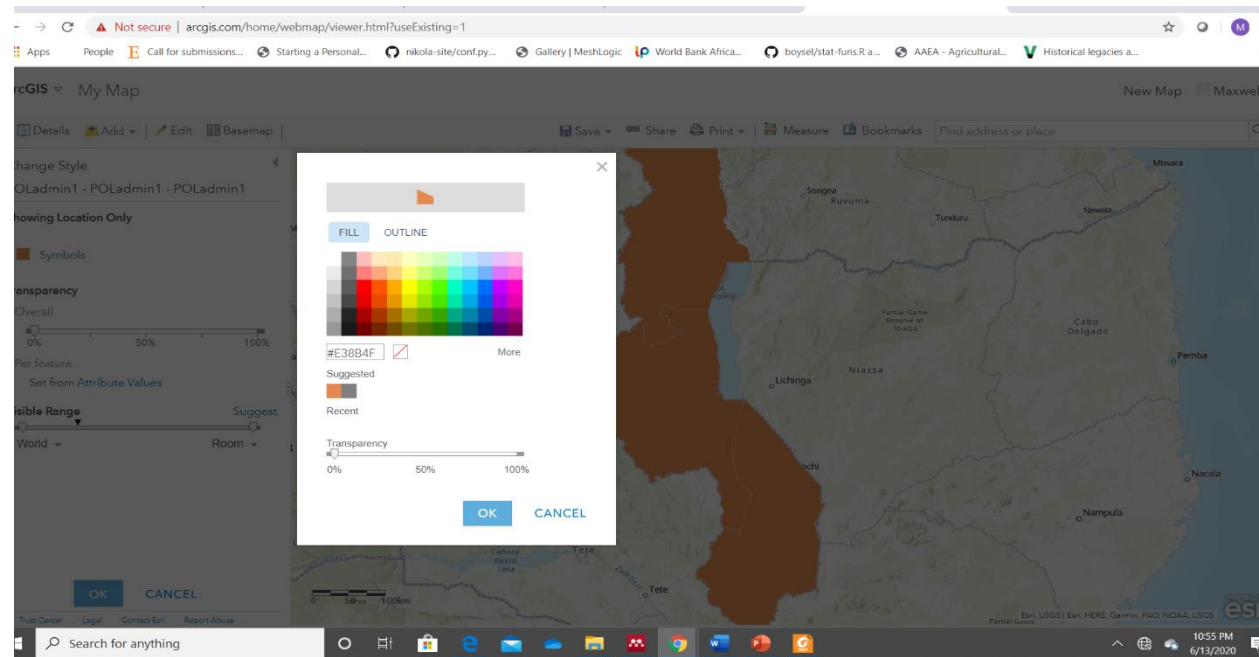
- Dynamic
- Easily shared
- Good for archiving data

Free version

The screenshot displays the ArcGIS website's home page in a web browser. The browser's address bar shows the URL `arcgis.com/home/index.html`. The page features a navigation bar with links for ArcGIS, Overview, Pricing, Map, Scene, Groups, and Content. A user profile for Maxwell Mkondiwa is visible in the top right corner. The main content area is dominated by a large blue banner with the ArcGIS logo and the Esri logo. Below the banner, there is a section titled "STORY MAPS LABS" which includes four featured story maps: "A Gallery of Amazing Species", "Crisis in Crimea", "Decisive Moments in the Battle of Gettysburg", and "Fighting Fires with GIS". At the bottom of the page, there are four columns of information: "Apps for Everyone" (Use ArcGIS apps to get the job done), "Make a Map" (Create a map that can be viewed in a browser, desktop or mobile device. Share it on a blog, via email, or embed it in a website.), "ArcGIS for Developers" (Build custom web and mobile applications that incorporate your maps and data.), and "Discover Lessons" (Take hands-on lessons based on real-world problems.). The Windows taskbar at the bottom shows the time as 10:43 PM on 6/13/2020.

Add shapefile/layer and change styles





Basemap



Imagery



**Imagery with
Labels**



Streets



Topographic



**Terrain with
Labels**



**Light Gray
Canvas**



**National
Geographic**



Oceans



OpenStreetMap

Qualitative data (categorical variables)

- Demo using categories of districts with more than 500, 000 people

Classifying Quantitative Data

- Demo by using household size in Esri Layer

Legend

- Configured in contexts tab
- Demo

Pop-Ups

- Data are pulled from attribute table
- User can configure and turn off field or turn off pop-ups entirely
- Links, images, and charts can be embedded

WIERD-ECON - Program x YUSUF CAN MASATLIOGLU x Pietro Ortoleva x solomon hsiang - Google Search x ArcGIS - My Map x

Not secure | arcgis.com/home/webmap/viewer.html?useExisting=1

Apps People Call for submissions... Starting a Personal... nikola-site/conf.py... Gallery | MeshLogic World Bank Africa... boysel/stat-funs.R a... AAEA - Agricultural... Historical legacies a...

ArcGIS My Map New Map Maxwell

Details Add Edit Basemap

Save Share Print Measure Bookmarks Find address or place

About Content Legend

Contents

- POladmin3
- POladmin1
- Topographic

+

-

Full Screen

REGION SOUTH

FEWS_ABB MI

FEWSID

FEWSID2 MI2CENTRAL

FEWSID3 MI3KASUNGU

Zoom to Edit

POladmin3 (Features: 171, Selected: 2)

AREA	ADMIN4_ID	ID	CNT_ABB	ADMIN2	ADMIN3	ADMIN4	NUM_CODE
0.05	35.00		MI	CENTRAL	KASUNGU	CHULU	
0.37	36.00		MI	CENTRAL	KASUNGU	KAOMBA	
0.07	37.00		MI	CENTRAL	KASUNGU	WIMBE	
0.11	38.00		MI	CENTRAL	KASUNGU	MALENGA CHANZI	
0.03	39.00		MI	CENTRAL	KASUNGU	KAREMBA	

True Center Legal Contact Us Report Abuse

Search for anything

11:03 PM 6/13/2020

Share the map

- Save your map
- Click the share button in the header menu
- Choose who can see the map
 - Everyone (public)
 - Members of a group
- Select a method of sharing
 - Link
 - Facebook
 - Embed in website (HTML)
 - Web application (JavaScript)
 - Mobile Devices (e.g., ArcGIS Collector)